H.N. HUY NGUYEN

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PROFILE

I'm a first-year PhD Student at the Laboratory of Semantic Analysis of Image and Text of Commissariat of Atomic Energy & Alternative Energy (CEA List) and Paris-Saclay University. My thesis focus on extraction and internalization of external knowledge for Vision-Language Models. I also contribute CEA's internal project for Robotics.

EXPERIENCES

- **PhD Student** *Jan 2026 - now*
  CEA List, Paris-Saclay University Palaiseau, Île-de-France
  - Internalization of external knowledge (e.g Knowledge Graphs) into foundation models of vision-language multi-modality.
  - Mitigate solutions to avoid catastrophic forgetting with continual learning methods.

- **AI Engineering Intern** *Feb - Aug 2025*
  Orange Innovation, Orange SA Châtillon, Île-de-France
  - Towards mastering the energy efficiency of Generative AI for code generation and summarization.
  - Studied of energy consumption of open-source LLMs and their alternatives such as post-training fine-tuned models and quantized models during inference.

- **Machine Learning Research Assistant Intern** *Apr - Aug 2024*
  Laboratoire Informatique Fondamentale et Appliquée de Tours (LIFAT EA 6300) Blois, France
  - Focused on Unsupervised learning on neural network development using Python/PyTorch for image reconstruction and image generation tasks.
  - Studied Autoencoders and Variational Autoencoders (VAEs) — latent generative models — implemented on the MNIST dataset.

FORMATION

- **INSA Centre Val de Loire** *2020 - 2025*
  French Grand Ecole - Industrial Engineering Blois, France
  Master of Engineering Acquisition, Analysis and Decision (ACAD)
  - Coursework includes: Machine Learning, Computer Vision, GPU Programming, Image Processing, Real-Time Systems, Embedded Systems, IoT, Electronics.

PROJECTS

- **Progressive Expansion of Variational Autoencoder** *Jan 2025*
  Tools : *Deep Learning, Python, PyTorch* []
  - Development and experimentation of a Variational Autoencoder (VAE) using a layer expansion approach to optimize performance.
  - Implementation of key techniques such as reparameterization trick and mini-batch gradient descent.

- **IoT Secure Access System with NFC/RFID and Face Recognition** *Nov 2024*
  Tools : *IoT, Raspberry Pi, POO, MySQL, Python* []
  - Designed and developed a secure access system using NFC/RFID badges and a low-resolution camera.
  - Implemented **dual authentication** combining badge usage and facial recognition.

SKILLS

- **Knowledge :** Machine Learning, Large Language Models, Computer Vision, IoT, HPC, Engineering, Electronics, Programming.
- **Programming languages :** Python, C/C++, docker, SLURM, shell script, Git & GitHub, MATLAB.
- **Frameworks & Libraries :** vLLM, Pytorch, Transformers, OpenCV, Numpy, Pandas, Matplotlib, tkinter, cuda, LaTeX.

LANGUAGES

English : Fluent | TOEIC 950/990 (11/2023), IELTS 7.0 (Aug 2022)
French : Professional working proficiency | DELF B2 (May 2022)
Vietnamese : Mother-tongue

Bourse & Certifications

- **Odon Vallet Scholarship** Organization Rencontres du Vietnam *2022*
- **DeepLearning.AI** Machine Learning Specialization *Sep 2023*
- **IBM AI Engineer Specialization** Introduction to Computer Vision and Image Processing *Feb 2024*
- **IBM AI Engineer Specialization** Deep Neural Networks with PyTorch *Jun 2024*
- **DeepLearning.AI** Neural Networks and Deep Learning *Jun 2024*

Activities & Interests

- **Extra curriculum** *Jul 2023*
  Produced an INSA campus discovery video for Vietnamese high school students.
- **Interests :** Running, Psychology, Nutrition & Cuisine, Stoicism, Neurosciences.